

Java Program: ClassCastException Handling

1. Aim

To write a Java program that demonstrates a ClassCastException by attempting to cast an object to an incompatible type and handles the exception using a try-catch block.

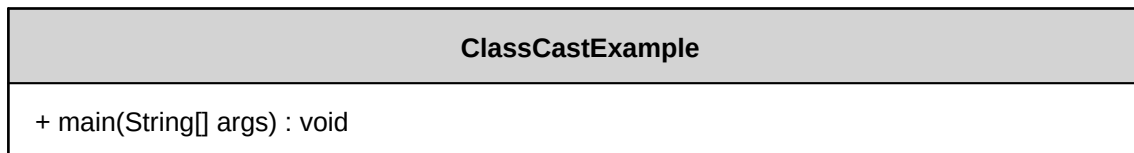
2. Description

A ClassCastException occurs when an object is cast to a class or type that it is not actually an instance of. In this program, an Object reference stores a String value. The program then attempts to cast that String object to Integer. Because a String object cannot be cast to Integer, Java throws a ClassCastException. The catch block handles the exception and displays an appropriate error message.

3. Algorithm

1. Start the program.
2. Enter the main method.
3. Create an Object reference and store a String value in it.
4. Attempt to cast the Object reference to Integer.
5. Since the actual object is a String, the cast is invalid and a ClassCastException is generated.
6. Catch the exception using a catch block.
7. Display a message indicating that an incompatible type cast was attempted.
8. Stop the program.

4. UML Diagram



5. Source Code with Comments

```
public class ClassCastException {  
    public static void main(String[] args) {  
        try {  
            // Store a String object in an Object reference  
            Object obj = "This is a string";  
  
            // Attempt to cast the String object to Integer  
            // This causes ClassCastException  
            Integer num = (Integer) obj;  
  
        } catch (ClassCastException e) {  
            // Handle the incompatible type casting exception  
            System.err.println(  
                "Caught ClassCastException: "  
                + "Incompatible type casting attempted."  
            );  
        }  
    }  
}
```

6. Compilation and Execution Commands

```
javac ClassCastException.java  
java ClassCastException
```

The first command compiles the Java source file. The second command executes the compiled program.

7. Output Screenshot

ClassCastException - Console Output

```
Caught ClassCastException: Incompatible type casting attempted.
```

8. Result

The Java program was successfully executed. It demonstrated a `ClassCastException` when a `String` object was incorrectly cast to `Integer`, and the exception was successfully handled using a try-catch block.